MeetSpace

AI-Powered Room Booking Assistant

SDAIA Academy · AI Productivity Bootcamp

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Final Capstone Project · 22 October 2026

Email and Excel make a simple booking a multi-step task

TASK BOUNDARIES

Start: requester sends a room-booking email.

End: coordinator records the reservation in Excel and sends confirmation.

WHY IT IS DIFFICULT

- Missing details require follow-up.
- Conflicts require checking alternatives.
- Booking is copied into a spreadsheet.

OBSERVED MANUAL BASELINE

Typical / complete	Messy / missing meeting time	Edge / unavailable room
2:38	**3:41**	**5:58**

Measured in a one-person educational simulation (requester and coordinator roles). Actual monthly frequency was not established.

Source: Productivity_Results.md · Final_Capstone_Board §1

Chosen track: Built Tool—not a standalone chat prompt

Why a tool is necessary

- A persistent room inventory and booking calendar.
- Capacity, availability, and maintenance rules.
- Safe conflict checks at the moment of confirmation.
- Booking, cancellation, and stored records.

React UI	Booking chat + calendars
AI model	Interprets booking requests
FastAPI	Validates and confirms
Firestore	Stores bookings + rooms

Live demo: https://meetspace-fucc.onrender.com/ | React + FastAPI deployed on Render; Firestore stores demo bookings.

The AI proposes; the user approves; the backend verifies

Before: email request → missing-information reply → Excel availability check → alternative email → Excel entry → confirmation email

STEP	NEW WORKFLOW	OWNER	CHECK
1	Enter date, time, attendees and room type	User	Request has required fields
2	Extract missing fields and offer suitable rooms	AI + backend	Capacity; time; conflicts; maintenance
3	Select a room and explicitly confirm	User	Correct room and time; consent
4	Recheck, save, show receipt and calendar	Backend + user	Atomic write; one booking; visible receipt

The model does not write directly to the database, and a suggested room is not a confirmed booking.

Source: AGENTS.md · SPEC workflow · booking tests

RICE prompt instructions define boundaries and acceptance

DELEGATED TO AI

Interpret Arabic/English requests, extract required fields, and ask for missing details.

DELEGATED TO BACKEND

Validate times, room type, capacity and availability; persist only after explicit confirmation.

RETAINED BY THE USER

Choose the room, approve the booking, review the receipt; approve product scope and changes.

Reusable prompt contract (RICE)

R / Role: bilingual booking parser

I / Instructions: never invent availability; ask for missing fields

C / Context: approved room inventory, current session and Riyadh time

E / Expected result: schema-validated JSON

Acceptance: no booking without consent; no confirmed double-booking in verified paths. Original operational instructions: AGENTS.md + backend/prompts/booking_agent.md.

Source: AGENTS.md · booking_agent.md · Prompt_Engineering_Log.md

Only fictional data were used; access remains a demo limitation

Identifiers removed?	No real employee records used; rooms and three demo identities were fictional from the start.
Institutional permission?	Not documented. Do not use real organizational data until policy and approval are verified.
Minimum data?	Booking date, time, room type, attendee count and demo identity; no meeting content required.
Never upload?	Firebase service-account files, .env, API keys, secrets, personal or classified meeting data.

Security limitation: demo employee/admin roles do not require secure login. Use fictional data only; real deployment needs authorization.

Three cases test completion, clarification and conflicts

CASE	INPUT	ACCEPTANCE RULE	OBSERVED EVIDENCE
Typical	Complete booking request	Suitable room; explicit consent; one saved reservation.	Live booking and persistence checks documented; timed user report 0:58.
Messy	Meeting time omitted in timed trial	Ask for missing time; retain known details; do not save early.	Agent requested clarification; timed user report 1:06.
Edge	Requested room conflicts with existing booking	Reject conflict; suggest alternatives; confirm only available option.	Conflict/concurrency checks documented; timed user report 1:46.

Verified quality evidence: 37 automated unit tests passed in the latest packaged run; measured bookings were not individually linked to Firestore IDs.

Source: TEST_CASE_EVIDENCE_MATRIX.md · M4 · Productivity_Results.md

Six functional checks: five passed; permissions partly met

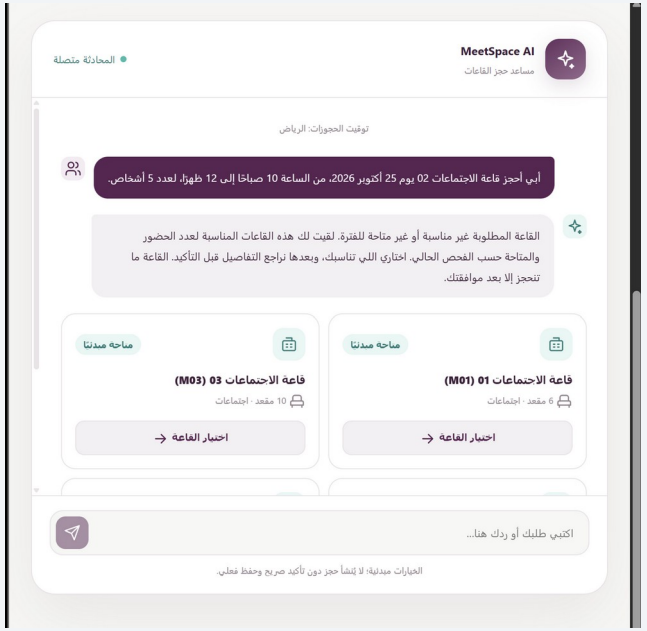
#	OFFICIAL CHECK	STATUS	WHAT THE EVIDENCE SHOWS
01	**Overlapping reservations**	**PASS**	Sara booked M02; Fahad's overlapping request was rejected; alternatives suggested.
02	**Invalid or missing inputs**	**PASS (4/4)**	Rejected past date, reversed time, overcapacity; asked for missing attendee count.
03	**Persistence and cancellation**	**PASS**	Reservation saved, survived browser refresh, then canceled successfully.
04	**Role permissions**	**PARTIAL**	Demo roles separated visually; no secure login or role authorization.
05	**Mobile experience**	**PASS**	Physical iPhone: booking chat, saved booking and calendar displayed and worked.
06	**Unavailable room**	**PASS**	M01 set to maintenance by admin; booking rejected; M02/M03 suggested.

Live demo: meetspace-fucc.onrender.com · Peer feedback not collected · Admin permissions PARTIAL (MVP limitation).

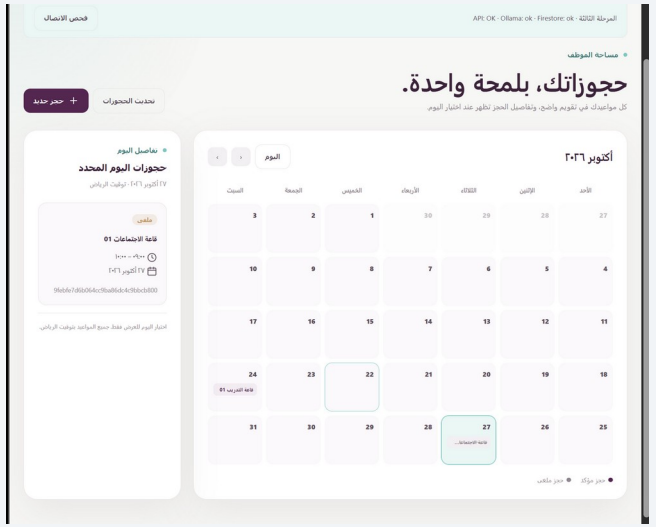
Source: user-run deployed demo tests, 22 Oct 2026 · SDAIA D4.3

Evidence from the deployed MeetSpace demonstration

Conflict handling and alternatives



Saved / canceled booking calendar



Real iPhone booking test

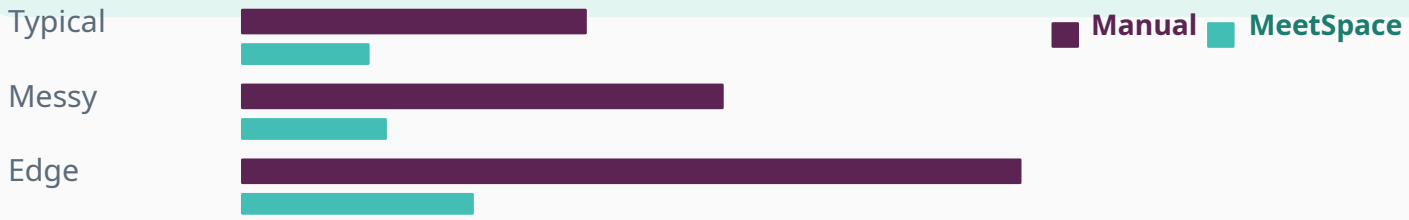


Live demo: https://meetspace-fucc.onrender.com/ | Captured from actual use of the deployed app; Arabic UI shown.

Source: screenshots of the Render deployment and iPhone test (22 Oct 2026)

68.79% less total elapsed time across three matched cases

SCENARIO	EMAIL + EXCEL	MEETSPACE	TIME SAVED
Typical	2:38	0:58	1:40
Messy	3:41	1:06	2:35
Edge	5:58	1:46	4:12
TOTAL	12:17	3:50	8:27



OBSERVED TOTAL REDUCTION

68.79%

737 seconds → 230 seconds

Matched scenarios, one timed run per workflow/case; one person simulated both manual roles. No invented email waiting. Active work, error counts and monthly frequency were not independently measured.

Source: Productivity_Results.md · productivity_results.csv

A real prompt regression was detected, rejected and corrected

A / BASELINE

Original prompt passed as benchmark baseline.

BASED ON EVIDENCE

B / REJECTED

A shorter RICE draft sometimes lost the supplied end time and mismatched language.

FAILED EXTRACTION

B2 / APPROVED

Restored three full bilingual JSON examples and tied language to the user text; passed 6 scenario trials / 8 turns.

OBSERVED RECOVERY

A later reduction in model calls was a backend optimization—not a prompt-only gain. Inference benchmarks are separate from the human before/after times.

Source: Prompt_Engineering_Log.md · M4 A/B/B2 benchmark files

Risks, human responsibility, AI disclosure and the 30-day plan

RISKS AND CONTROLS

False booking success

The backend confirms only after a verified Firestore save.

Wrong room or conflict

Capacity, maintenance and overlaps are rechecked before commit.

Unauthorized access

Demo roles are switchable without secure login; keep fictional data only until authorization is implemented.

PLANNED ADOPTION — NOT YET EXECUTED

DAY 1 — Verify the deployed link and confirm the instructor accepts slide-only submission.

WEEK 1 — Record a peer usability check and track booking failures or corrections.

DAY 30 — Review usage evidence and access controls; continue, revise, or stop.

AI USE DISCLOSURE

AI tools (including Codex) assisted development; the author directed, tested and reviewed the final work.

Open items: independent peer feedback not collected; secure role access not implemented; official live-demo alternative unconfirmed.

Source: Final_Capstone_Board §8 · EVIDENCE_GAPS.md